

Losing a home in a wildfire costs more than just the rebuild

Lessons from “The Distribution and Consequences of Natural Disasters: Evidence from U.S. Wildfire Victims” by Patrick Baylis, Judson Boomhower, Jonathan Colmer, and John Voorheis, NBER Working Paper, 2026.

New research studying around 150 major U.S. wildfires over two decades shows that **when a wildfire destroys a home, occupants don’t just lose the structure — they lose income too, for years afterward.**

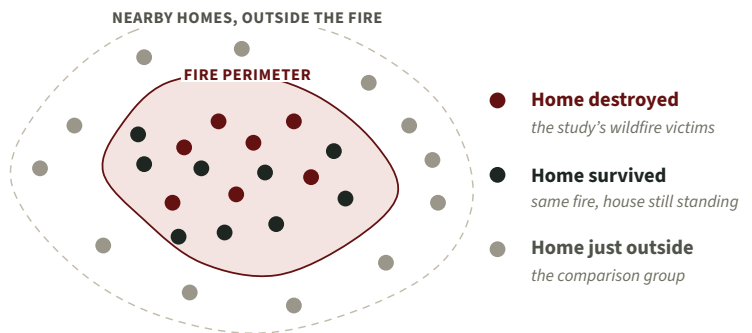
BACKGROUND

Global economic losses from natural catastrophes totaled \$320 billion in 2024. Though this number is striking, it masks that **the financial consequences of disasters are unevenly distributed across households.** To respond well to disasters, it’s important to know who bears the losses, whether the damage to household finances goes beyond the cost of rebuilding, and what happens to a neighborhood as it recovers. This is especially true for areas prone to wildfires, which are increasing in frequency and severity.

In a new working paper, Baylis et al. (2026) find that those whose homes were destroyed in a wildfire

don’t just lose the value of their house, they also lose a significant portion of their income for years afterward. The researchers studied around 150 major wildfires in five western states between 2003 and 2022 — the fires behind more than 70% of all U.S. homes destroyed by wildfire in that period — and analyzed property damage alongside occupant demographics and income data.

WHO THE STUDY COMPARES



Illustrative schematic — not a map

THE CHALLENGE

Measuring what a wildfire costs the people in its path is harder than it looks. Comparing a household’s income before and after the fire risks crediting the fire with changes already underway in the local economy.

Comparing wildfire victims to people elsewhere raises a different concern: fires strike particular places, and the households in their path were already wealthier and older than the country as a whole. Narrowing to a single fire — setting households whose homes burned against neighbors whose homes came through standing — puts both groups in the same place at the same moment. Homes did not survive at equal rates, however: those that burned belonged to lower-income households than those left standing, so the two groups differed before the fire started.

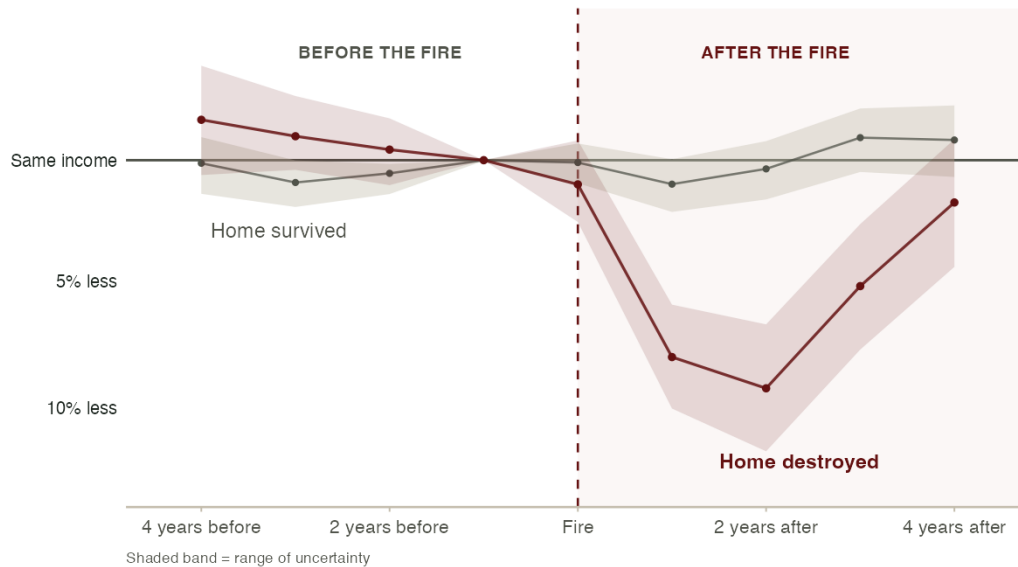
Baylis et al. instead tracked the two groups separately, comparing each against people living just outside the fire’s perimeter. These neighbors shared the same local job market, so whatever moved incomes across the area moved them for everyone. Before each fire, all three groups’ incomes rose and fell together, supporting their use as a stand-in for what victims would have earned had the fire never reached them.

INCOME FELL ONLY FOR THOSE WHO LOST A HOME

Households whose homes were destroyed earned less than their neighbors just outside the perimeter for three years running — about 11% less at the low point in the second year, before catching back up by the fourth (see figure). Added up, the shortfall comes to roughly 30% of a single year’s pre-fire income, or about \$38,000, on top of what the rebuild itself cost. Households whose homes survived the same fires barely departed from those neighbors at all. **The loss tracks whose house burned, not who lived through the fire.**

THE WILDFIRE'S EFFECT ON INCOME, YEAR BY YEAR

Source: Baylis, Boomhower, Colmer & Voorheis (2026), Fig. 4a.



Note: This figure shows the estimated effect of a wildfire on the incomes of people who lived inside its perimeter. One line follows occupants whose homes were destroyed; the other follows occupants of homes in the same fires that survived. Both are measured against similar people who lived just outside the perimeter, and both are set to zero in the year before the fire. A point below the line means that group had fallen behind those outside neighbors. The estimates account for occupants' characteristics and for anything that affected everyone caught in the same fire in the same year. If living through a fire, rather than losing a home to the fire, is what costs people income, both of the lines would fall after the fire.

IMPLICATIONS FOR POLICY

Wildfire loss in the U.S. is often pictured as a wealthy Californian's problem. These households were better off than the country as a whole on average, though within a single fire it was the lower-income homes that burned — and a quarter of the people who lost one earned under \$42,000.

Four years after a fire, half of destroyed addresses are still unoccupied, and the homes that do come back tend to house higher-income, younger residents than those who lived there before. **Recovery aid measured by how fast a neighborhood rebuilds can miss whether the people who lost their homes are the ones who return.**

This is one set of large U.S. wildfires between 2003 and 2022, and the paper does not settle why income falls. What it does establish is the size of the loss: enough that **helping households protect a home before a fire may be worth more than it appears when only the structure is counted.**

FOR MORE INFORMATION

Contact [Corresponding author] ([email]) with questions about this research.

Baylis, Patrick, Judson Boomhower, Jonathan Colmer, and John Voorheis. 2026. "The Distribution and Consequences of Natural Disasters: Evidence from U.S. Wildfire Victims." NBER Working Paper. [nber.org/papers/... — add once available].

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